Truminds AI Skill Library

Architecture and end-to-end flow · team walkthrough

What we are building

Small reusable AI skills, chained into workflows and packaged as pods, with a quality check after every step

1

Skill

One small AI capability, e.g. requirement review, design, unit tests. Technology- and domain-agnostic.

2

Workflow

Skills in order with human gates, e.g. brownfield-change: requirement → design → code → PR.

3

Pod

A ready-made AI-augmented team, e.g. 1 Solution Architect + 1 Full-stack + 1 QA agent.

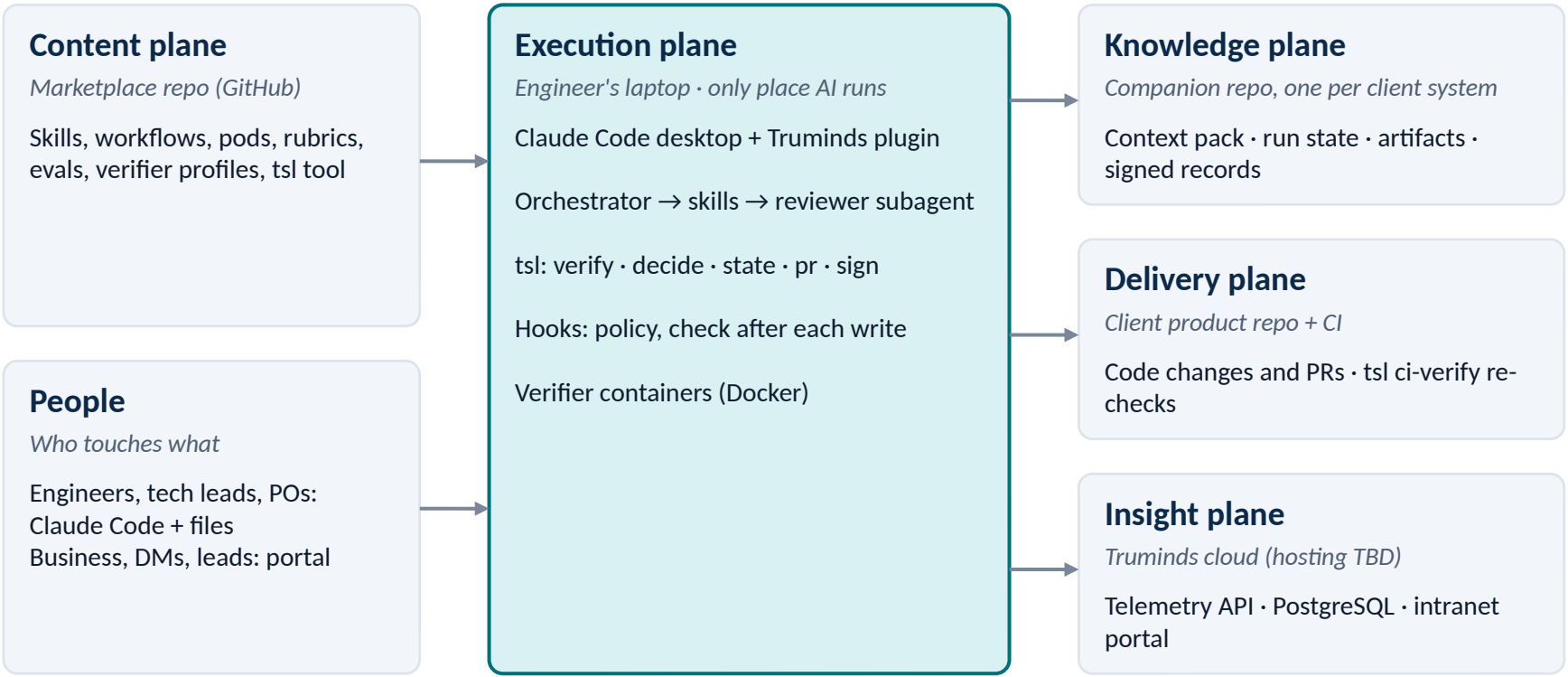
4

Harness

Checks every output against the client system, scores confidence, stops for a human when needed.

- Runs locally in Claude Code desktop + Truminds plugin
- Any domain, greenfield or brownfield
- Company Claude seats only · no Claude API
- People decide; code makes pass/fail decisions

Core components: five planes, AI runs in only one



Inside one step: the AI proposes, code decides

1	**/truminds:run**	Engineer starts a workflow on a ticket; tsl takes a lock and creates the run
2	**Skill writes artifact**	e.g. design.md, with Known / Assumed / Unknown and cited claims
3	**Hook → tsl verify**	Code checks facts: schema, sections, names exist in the context pack, tests pass
4	**Reviewer subagent**	Checks meaning against the context pack; every finding must cite evidence
5	**tsl decide**	Code verifies citations, computes band, picks the outcome, signs the record

Six outcomes

- **PASS** — next step
- **PASS WITH FINDINGS** — continue, notes to PR
- **RETRY** — AI fixes, max 2–3
- **NEEDS INPUT** — questions file, pause
- **ESCALATE** — tech lead decides
- **ABORT** — secret, PII, blocked action

Where things live and what crosses each boundary

Laptop

Claude Code + plugin, tsl, hooks, Docker verifiers, offline outbox

Client code is read here, on company seats

Companion repo

tsl.yaml, context pack, run state, open questions, approvals, artifacts and their signed check records

Truminds-owned, one per client system

Product repo

Code changes and PRs only; CI runs tsl ci-verify

Client-owned; nothing else is written here

Cloud

Telemetry API, PostgreSQL, intranet portal

Metadata only: ids, counts, outcomes, durations. No code, prompts or client names

Trust: every record is signed with the engineer's key (registered at tsl login). CI re-runs the code checks and verifies signatures; a code PR without records fails.

End to end: seven steps from ticket to merged PR



1	2	3	4	5	6	7
Onboard system	**Pick workflow / pod**	**Run in Claude Code**	**Harness checks**	**Human gates**	**Pull request**	**Insights**
Tech lead builds and signs context pack	Plugin installed from GitHub	Skills run one after another	After every step	Questions, approvals	Code + tests + records	Metadata → portal, skills improve

Pause and resume

When a step needs a decision, tsl writes runs/<run_id>/open-questions.md and the run pauses (Claude session can close). The PO answers in the file; the engineer runs /truminds:resume <run_id>. Approvals work the same way through approvals.md and tsl approve.

Example: ABC-142, show 30 days of order history

Brownfield change on an existing order service · workflow brownfield-change · engineer runs /truminds:run brownfield-change ABC-142

1	Requirement review	PASS · High	Requirement is clear; terms match the glossary
2	Story split	PASS WITH FINDINGS · Medium	Time-zone edge case noted for the PR
3	Impact analysis	RETRY → PASS · High	First try named a table that does not exist; code check caught it, AI fixed it
4	Design	NEEDS INPUT	Reviewer cites data-model.md: history is kept 24 hours, not 30 days. Q-01 goes to the product owner; run pauses

Example continued: from the answer to a merged PR

5 PO answers Q-01

Option A: extend retention to 30 days

6 /truminds:resume

Design re-runs with the answer → PASS

7 Design gate

Tech lead approves (tsl approve, signed)

8 Threat model

STRIDE rows for each component

9 Implement + unit tests

14 of 14 checks · coverage up

10 Secure review + test plan

Every acceptance criterion has a test

11 tsl pr

PR with code, test results, records

12 CI + reviewer

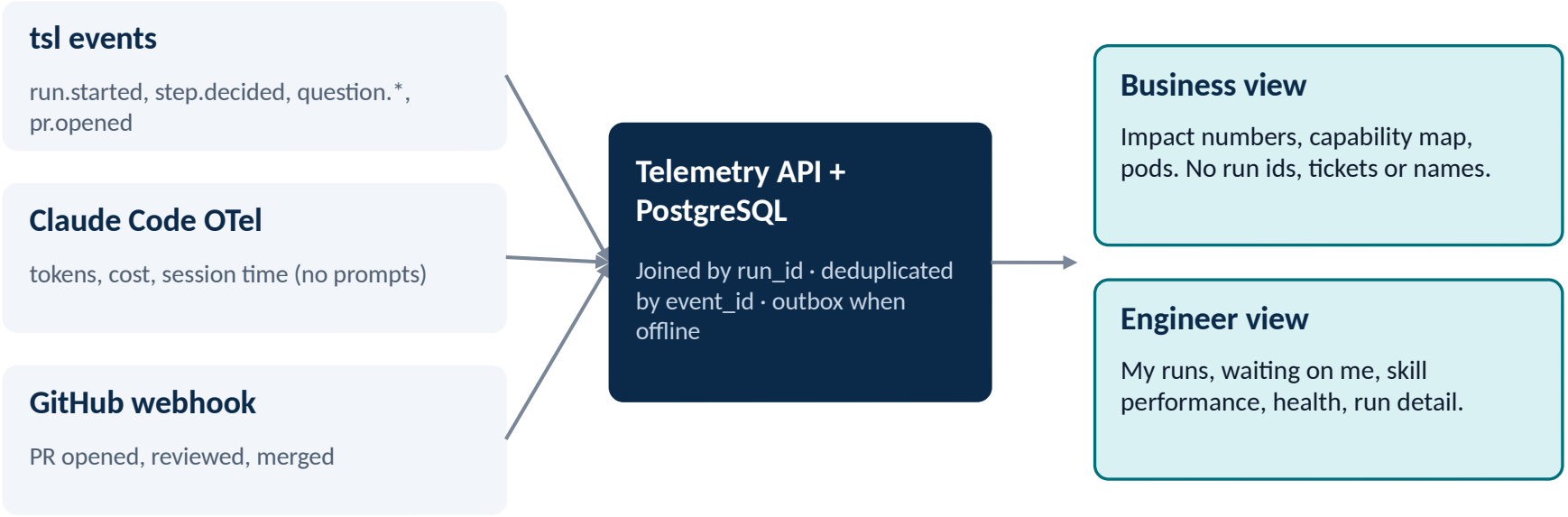
ci-verify re-checks; a human merges

13 Insights

Events update dashboards; skills learn

Amber boxes are human decisions

How results reach the portal



Self-improvement: a weekly report (plain code) shows each skill's failure causes; the owner adds a failing eval case first, then improves the skill with /truminds:improve.

Next steps: who owns what

Owner	Owns	First deliverable
Abhishek Backend	Run state and locks, /truminds run · resume · status, telemetry API + PostgreSQL	State machine + run-state schema; events API skeleton with idempotency
Shubham Backend	Plugin + marketplace, hooks and policy, tsl pr, ci-verify action, verifier containers, OTel + webhooks	Plugin installs from marketplace; a hook calls tsl verify on write
Japman Backend	Harness: tsl verify / decide, rubrics, citation check, evals, context extractors	decide() golden-file tests with H1–H7 fixtures; init-context on a sample system
Ekjot Frontend	Portal: catalogue, skill detail, pods, capability map, get started	Catalogue from marketplace data; recommendation on Backstage vs custom web app
Gaurav Frontend	Home (Engineer), insights dashboards, run detail, question / PR / approval templates	Dashboards on mock events; templates validated against schemas